

A NEW ESTIMATION PROCEDURE BASED ON FREQUENCY MOMENTS

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Summary: The paper considers applications of a continuous analogue of the method of frequency moments used by Sichel. A general estimation procedure is proposed for scale-related parameters of continuous distributions based on equating the log-likelihood function with $-n\pi_X(s, \theta)$, where $\pi_X(s, \theta) = s^{-1} \log \int f_X^{s+1}(x, \theta) dx$ is a frequency-related function which approaches the entropy parameter as $s \rightarrow 0$. This limiting case is satisfied at the maximum likelihood solution in exponential families, providing a hitherto unknown equation satisfied by the maximum likelihood estimator. In distributions not belonging to this family, the method provides an alternative estimator whose use is demonstrated in the estimation of scale in a Cauchy distribution, where traditional moments fail. The procedure provides a class of estimators which bound maximum likelihood with arbitrary accuracy in exponential families, without the use of calculus. Other examples are taken from the normal, gamma and Pareto distributions.

1. Introduction

Frequency moments, i.e. moments of $Y = f_X(X)$, where f_X denotes the density function of the random variable X , were first introduced into parameter estimation by Yule (1938) on a suggestion of Karl Pearson. This parametric

method of estimation has not been extensively used in the literature, with the notable exception of Sichel (1947, 1949, 1982) who applies it with good effect to several distributions, including the lognormal, Type VII and the distribution for which he is renowned, the generalised inverse Gaussian Poisson.

We briefly describe Sichel's procedure. Suppose that $p_X(x; \theta)$ denotes the probability distribution of a discrete X which depends on the unknown parameter θ . Let

$$\omega_X(s, \theta) = E(Y^s) = \sum_x p_X^{s+1}(x, \theta),$$

where s is some real value for which $E(Y^s)$ exists, as well as being the argument of the so-called 'frequency moment' function, ω_X . The method proposes an estimating equation obtained by equating the moment, $\omega_X(s, \theta)$, with its empirical counterpart for given s , giving

$$\omega_X(s, \hat{\theta}) = \sum \hat{p}_x^{s+1}, \quad (1)$$

where a random sample of n observations of X yields $n\hat{p}_x$ observations of the type $\{X = x\}$.

In order to use this method in continuous distributions, Sichel (1967) suggests "splitting the variate range into equidistant class intervals and applying a suitable correction for grouping". He also notes that a disadvantage of the method of frequency moments "is to be found in its incapability of estimating parameters of location per se ... [and] has to borrow from the methods of moments or quantiles." This is due to the fact that $\omega_X(s, \theta) = \omega_{X-\mu}(s, \theta)$ for any μ . It is thus assumed that θ is not a location parameter. In continuous distributions, this approach is also disadvantaged by two facts: (a) relatively

large sample sizes are required for grouping, and (b) the grouping itself must be corrected for in the estimation. This is the result of the discrete procedure applied to discretised continuous data.

Yule lets $s = 1, 2$ and 3 while Sichel lets $s = 0.5$ and 1 , both noting that this method is best when s is close to 0 . For $s = 0$, (1) clearly fails. However, by taking the s th root of (1) and letting $s \rightarrow 0$ we obtain an estimating equation based on the observed entropy of the sample, $-\sum \hat{p}_n \log \hat{p}_n$, which appears to be highly efficient under certain circumstances.

In the present paper, a continuous analogue of the method of frequency moments used by Sichel is proposed, based on frequency values corresponding to a random sample from the population. The approach mirrors the method of moments procedure based on the transform $Y = f_X(X)$. The proposed analogue uses exact frequency measures in the estimation, thus removing the problems of grouping in Sichel's procedure. In particular, it is applicable to small sample sizes without a correction factor.

In the next section, we derive the estimator and its limiting case. The method is then examined in a one-parameter exponential family of distributions where it is shown that the limiting case yields the maximum likelihood estimator (MLE). In the multi-parameter exponential family, the method gives a combination of the likelihood estimates, providing a hitherto unknown 'likelihood equation'. The class also provides estimates which bound the maximum likelihood solution for this family. Examples illustrating this estimator are given for the normal, gamma, Cauchy and Pareto distributions, and further applications are suggested.

2. The estimation procedure based on frequency moments

In traditional moment estimation, the values of the x -observations are used to infer unknown parameter values. In particular, the procedure is based on the premise that if x and y are independent observations from densities $f_X(\cdot, \mu_X)$ and $f_Y(\cdot, \mu_Y)$, with respective means μ_X and μ_Y and we observe $x < y$ then we infer that $\mu_X < \mu_Y$.

In a similar way, the density, $f_X(x, \theta)$ at $X = x$ imparts information about the scale of X . In particular, if x and y are independent observations from $f_X(\cdot, \sigma_X)$ and $f_Y(\cdot, \sigma_Y)$, where σ_X and σ_Y denote the respective standard deviations, say, and $f_X(x, \sigma_X) > f_Y(y, \sigma_Y)$, then it is reasonable to expect that $\sigma_X < \sigma_Y$.

In ordinary moment estimation we equate the t th power of the observation, x , with its expectation $E(X^t)$ for given t (usually $t = 1$ or 2). Analogously, the proposed procedure suggests equating the s th power of $Y = f_X(X, \theta)$ with its expected value to give

$$\omega_X(s, \hat{\theta}) = f_X^s(x, \hat{\theta}),$$

for given s , where $\omega_X(s, \theta) = \int f_X^{s+1}(x, \theta) dx$. Upon taking logarithms this becomes

$$\pi(s, \hat{\theta}) = \log f_X(x, \hat{\theta}), \quad (2)$$

where $\pi_X(s, \theta) = \{\log \omega_X(s, \theta)\}/s$.

Now consider a random sample, $x_1, \dots, x_n$ of X . For each x_i , (2) gives $\pi_X(s, \hat{\theta}_i) = \log f_X(x_i, \hat{\theta}_i)$, where $\hat{\theta}_i$ denotes the estimate of θ using x_i , $i = 1, \dots, n$. To obtain a single estimator based on the entire sample, we

average these estimators over $i = 1, \dots, n$ and replace each $\hat{\theta}_i$ by the single estimator $\hat{\theta}$ to obtain

$$\pi_X(s, \hat{\theta}) = \frac{1}{n} \ell_X(\hat{\theta}), \quad (3)$$

where $\ell_X(\theta) = \log \Pi_i f_X(x_i, \theta)$ is the log-likelihood function. The space of estimators of the type (3) over all s for which ω_X exists, defines the proposed class.

An interesting case arises in the limit as $s \rightarrow 0$. From L'Hospital's rule it follows that

$$\begin{aligned} \lim_{s \rightarrow 0} \pi_X(s, \theta) &= \lim_{s \rightarrow 0} \frac{1}{s} \log E(Y^s) \\ &= \lim_{s \rightarrow 0} E(Y^s \log Y) / E(Y^s) \\ &= E(\log Y) = -H_X(\theta), \end{aligned}$$

where $H_X(\theta)$ is the so-called 'entropy' parameter of X , and is defined as $H_X(\theta) = -\int f_X(x, \theta) \log f_X(x, \theta) dx$. Entropy was introduced into the literature by Shannon (1948) and is used as an index of diversity in population distributions; most of its applications are in information theory (see e.g. Kotz, 1966).

The limiting case of (3) as $s \rightarrow 0$ is thus

$$\ell_X(\hat{\theta}) = -nH_X(\hat{\theta}), \quad (4)$$

which is, in effect, the first moment estimator based on the transformation $Y = \log f_X(X_i)$. Clearly (4) is consistent since $\ell_X(\theta)/n = \sum \log f(x_i, \theta)/n \rightarrow E(\log f_X(X, \theta)) = -H_X(\theta)$, almost surely, as $n \rightarrow \infty$.

3. Exponential families

Following Lehmann (1959), we define X as belonging to the one-parameter exponential family of continuous distributions if its density function can be

expressed in the form

$$f_X(x, \theta) = e^{\theta T(x) - A(\theta)},$$

over an x -range which is independent of θ . It is well known that $\sum T(x_i)$ is minimal, complete and sufficient for θ in this class. It is easily shown that

$$\pi_X(s, \theta) = \frac{A((s+1)\theta) - A(\theta)}{s} - A(\theta),$$

so that the entropy of X may be expressed as

$$H_X(\theta) = -\lim_{s \rightarrow 0} \pi_X(s, \theta) = A(\theta) - \theta A'(\theta).$$

Since the log-likelihood is

$$\ell_X(\theta) = \theta \sum_{i=1}^n T(x_i) - nA(\theta),$$

it follows that the estimator, (3) becomes

$$\frac{1}{n} \sum_{i=1}^n T(x_i) = \frac{A((s+1)\hat{\theta}) - A(\hat{\theta})}{s\hat{\theta}}. \quad (5)$$

From the definition of the derivative, it follows that as $s \rightarrow 0$, (5) becomes

$$A'(\hat{\theta}) = \frac{1}{n} \sum_{i=1}^n T(x_i). \quad (6)$$

Differentiating both sides of $\int e^{\theta T(x)} dx = e^{A(\theta)}$ with respect to θ gives $E\{T(X)\} = A'(\theta)$, indicating that this is the first moment estimator based on the transform, $T(X)$. A second differentiation gives $\text{var}\{T(X)\} = A''(\theta) > 0$, so that $A'(\theta)$ is an increasing function of θ . In addition, it follows from general MVUE theory that the only transformations of θ whose MVUEs attain the Cramér-Rao bound are linear transformations of $A'(\theta)$.

Under the regularity conditions, it is a simple matter to show that (6) is equivalent to the maximum likelihood estimator of θ by equating the first

derivative of $\ell_X(\theta)$ to 0. In addition, it is known from Jensen's inequality that $\{E(Y^s)\}^{1/s} \leq \{E(Y^t)\}^{1/t}$ for $s < t$, (see e.g. Moran, 1968, p.234) from which it follows that $\pi_X(s, \theta)$ is an increasing function of s . Since $A'(\theta)$ is increasing in θ , it follows that two estimates obtained from (5) with positive and negative s respectively, will bound the maximum likelihood estimator. These bounds can be made arbitrarily accurate by choosing the s -values sufficiently close to 0.

4. The multi-parameter case in exponential families

The single parameter case is easily generalised to a vector of unknown parameters, $\theta = (\theta_1, \dots, \theta_k)$ with X belonging to the k -parameter exponential family of distributions, whose density function is of the form

$$f_X(x, \theta) = \exp \left\{ \sum_{j=1}^k \theta_j T_j(x) - A(\theta) \right\}.$$

The log-likelihood is

$$\ell_X(\theta) = n \sum_{j=1}^k \theta_j \bar{T}_j - nA(\theta),$$

where $\bar{T}_j = \sum_{i=1}^n T_j(x_i)/n$, $j = 1, \dots, k$. The estimator, (3), becomes

$$\sum_{j=1}^k \hat{\theta}_j \bar{T}_j = \frac{A((s+1)\hat{\theta}) - A(\hat{\theta})}{s} \quad (7)$$

where $(s+1)\theta = ((s+1)\theta_1, \dots, (s+1)\theta_k)$. As in the single-parameter case, the statistics employed are complete and minimally sufficient.

To obtain the limiting case of (7) as $s \rightarrow 0$, we note that

$$\pi_X(s, \theta) = \frac{A((s+1)\theta) - A(\theta)}{s} - A(\theta)$$

and

$$-H_X(\theta) = \sum_{j=1}^k \theta_j E\{T_j(X)\} - A(\theta) = \sum_{j=1}^k \theta_j \frac{\partial}{\partial \theta_j} A(\theta) - A(\theta).$$

Here $E\{T_j(X)\} = \partial A(\theta) / \partial \theta_j$ is again obtained by differentiating both sides of the identity $\int f_X(x) dx = 1$ with respect to θ_j , $j = 1, \dots, k$. Since $\pi_X(s, \theta) \rightarrow -H_X(\theta)$ as $s \rightarrow 0$, it follows that

$$\lim_{s \rightarrow 0} \frac{A((s+1)\theta) - A(\theta)}{s} = \sum_{j=1}^k \theta_j \frac{\partial}{\partial \theta_j} A(\theta).$$

This is a well-known result in the theory of directional derivatives (see e.g. Edwards, 1973, p.69), derived here by alternative considerations. As $s \rightarrow 0$, (7) thus becomes

$$\sum_{j=1}^k \hat{\theta}_j \frac{\partial}{\partial \theta_j} A(\hat{\theta}) = \sum_{j=1}^k \hat{\theta}_j \bar{T}_j, \quad (8)$$

which is the linear combination of the likelihood equations

$$\frac{\partial}{\partial \theta_j} A(\hat{\theta}) = \bar{T}_j, \quad (j = 1, \dots, k).$$

Equation (8) is obtained by multiplying each likelihood equation by the parameter in the partial derivative and summing. Its form permits an alternate derivation of the likelihood equations in this important class.

5. Examples

5.1 Scale parameters

If the distribution has a single scale parameter, θ , (i.e. the distribution of $Z = X/\theta$ is independent of θ) then

$$\pi_X(s, \theta) = -\log \theta + \pi_Z(s),$$

since $\omega_X(s, \theta) = \theta^{-s} \omega_Z(s)$. The class (3) becomes

$$\pi_Z(s) = \frac{1}{n} \sum_{i=1}^n \log f_Z(x_i/\hat{\theta}),$$

which in the limit as $s \rightarrow 0$ is

$$-H_Z = \frac{1}{n} \sum_{i=1}^n \log f_Z(x_i/\hat{\theta}),$$

where $\pi_Z(s)$, f_Z , and H_Z are independent of θ .

5.2 Normal distribution

Suppose the density of X is

$$f_X(x, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\{(x-\mu)/\sigma\}^2}, \quad (-\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0),$$

where μ is assumed known. This is an established member of the one-parameter exponential family.

The estimator of σ^2 derived from (3) is

$$\hat{\sigma}^2 = \frac{s}{\log(1+s)} \sum_{i=1}^n (x_i - \mu)^2/n,$$

the familiar sum of squares multiplied by the factor $1/h(s) = s/\log(1+s)$, where $\lim_{s \rightarrow 0} h(s) = 1$. For unbiasedness of $\hat{\sigma}^2$ when μ is estimated by the

sample mean, $\bar{x}$, we require that

$$h(s) = 1 - \frac{1}{n}. \quad (9)$$

Table 1 below provides for values of n and s satisfying (9).

Table 1. Values of (s, n) for which frequency moment estimates provide the MVUE in normal and exponential distributions.

n	s
1	∞
2	2.5130
5	0.5386
10	0.2302
20	0.1071
50	0.0411
100	0.0203
200	0.0101
500	0.0040

5.3 Gamma distribution

Consider the two-parameter gamma distribution with density function

$$f_X(x, r, \lambda) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}, \quad (x > 0, \lambda > 0, r > 1), \quad (10)$$

a member of the two-parameter exponential family, where in the above notation $r - 1 = \theta_1$, $\theta_2 = \lambda$, $T_1(x) = \log x$, $T_2(x) = -x$ and $A(\theta) = \log \Gamma(\theta_1 + 1)$.

Thus (7) becomes

$$(\hat{r} - 1) \log(\hat{\lambda} \bar{g}) - \hat{\lambda} \bar{x} = \frac{1}{s} \left\{ \log \frac{\Gamma(s(\hat{r} - 1) + \hat{r})}{\Gamma(\hat{r})} - \hat{r} \log(1 + s) \right\} - (\hat{r} - 1) \log(1 + s), \quad (11)$$

where $\bar{x} = \sum x_i/n$ and $\bar{g} = \Pi x_i^{1/n}$, the sample arithmetic and geometric means, respectively. In the limit as $s \rightarrow 0$, this becomes

$$(\hat{r} - 1) \log(\hat{\lambda}\bar{g}) = (\hat{r} - 1)\psi(\hat{r}) - \hat{r}, \quad (12)$$

where $\psi(r) = \Gamma'(r)/\Gamma(r)$ denotes the digamma (psi) function. Comparing the form of (8) with (12) yields the likelihood equations

$$\log(\hat{\lambda}\bar{g}) = \psi(\hat{r}) \quad \text{and} \quad \hat{\lambda}\bar{x} = \hat{r}.$$

Putting $\hat{\lambda} = \hat{r}/\bar{x}$ into (11) leads to the estimator of r alone:

$$(\hat{r} - 1) \log(\hat{r}\bar{g}/\bar{x}) - \hat{r} = \frac{1}{s} \left\{ \log \frac{\Gamma(s(\hat{r} - 1) + \hat{r})}{\Gamma(\hat{r})} - \hat{r} \log(1 + s) \right\} - (\hat{r} - 1) \log(1 + s), \quad (13)$$

which reduces to the likelihood equation $\log(\hat{r}\bar{g}/\bar{x}) = \psi(\hat{r})$ as $s \rightarrow 0$.

From section 4, an estimator for r which shares the asymptotic properties of maximum likelihood can be found by choosing s in (13) so that $s \rightarrow 0$ as $n \rightarrow \infty$. An obvious choice is $s = 1/\{n(\hat{r} - 1)\}$, for which (13) becomes

$$\log(\hat{r}\bar{g}/\bar{x}) - \frac{\hat{r}}{\hat{r} - 1} = n \log \frac{\Gamma(\hat{r} + n^{-1})}{\Gamma(\hat{r})} - (n\hat{r} + 1) \log \left\{ 1 + \frac{1}{n(\hat{r} - 1)} \right\}. \quad (14)$$

Substituting the first-order approximation $\log(1 + q) \approx q$ into (14) with $q = \{n(\hat{r} - 1)\}^{-1}$ yields the computationally simpler form

$$\log(\hat{r}\bar{g}/\bar{x}) = n \log \frac{\Gamma(\hat{r} + n^{-1})}{\Gamma(\hat{r})} - \frac{1}{n(\hat{r} - 1)}, \quad (15)$$

best applied for large n . It is clear that both (14) and (15) approach the likelihood equation, $\log(\hat{r}\bar{g}/\bar{x}) = \psi(\hat{r})$ as $n \rightarrow \infty$. Estimators which converge faster toward the likelihood solution may be obtained from (13) by letting

$s = n^{-2}(\hat{r} - 1)^{-1}$, say, although the formulation of these becomes more complex.

To provide a numerical illustration of the rates of convergence of these estimators to the MLE in a hypothetical case, let $\bar{x} = 5.0000$ and $\bar{g} = 4.5091$ so that the MLE of r is $\hat{r}_{ml} = 4.9991$. Table 2 shows estimated values of r from (14) and (15) for $n = 10, 20, 50, 100, 200, 500, 1000$.

Table 2. Rate of coverage of frequency moment estimates to MLE ($\hat{r} = 4.9991$) in a gamma distribution with $\bar{x} = 5$ and $\bar{g} = 4.5091$.

n	$\hat{r}$ from (14)	$\hat{r}$ from (15)
10	4.9191	5.6318
20	4.9592	5.3201
50	4.9832	5.1288
100	4.9912	5.0642
200	4.9951	5.0317
500	4.9945	5.0121
1000	4.9983	5.0056

The important special case of the exponential distribution is obtained when $\hat{r} = 1$, when (11) reduces to

$$\hat{\lambda} = \frac{1}{\bar{x}} h(s), \quad (16)$$

the familiar MLE multiplied by the factor, $h(s)$, obtained in the normal distribution. For (16) to provide the MVUE of λ , the value of s must again be chosen so that (9) holds (see table 1).

5.4 Cauchy distribution

Suppose that the density function is

$$f_X(x, \theta) = \frac{1}{\pi\theta\{1 + (x/\theta)^2\}}, \quad (-\infty < x < \infty, \theta > 0).$$

Here θ is a pure scale parameter so that the conditions of 5.1 apply. It is, however, not a member of the exponential family. For this case, it is easily shown that

$$\pi_Z(s) = \frac{1}{s} \left\{ \log B\left(s + \frac{1}{2}, \frac{1}{2}\right) - \log \pi \right\} - \log \pi,$$

so that

$$\begin{aligned} \lim_{s \rightarrow 0} \pi_Z(s) &= -\log \pi + \frac{d}{ds} \log B\left(s + \frac{1}{2}, \frac{1}{2}\right) \Big|_{s=0} \\ &= -\log \pi + \frac{1}{\sqrt{\pi}} \frac{d}{ds} \frac{\Gamma(s + \frac{1}{2})}{\Gamma(s + 1)} \Big|_{s=0} \\ &= -\log \pi + \frac{1}{\sqrt{\pi}} \left\{ \Gamma'(\tfrac{1}{2}) - \Gamma'(1)\Gamma(\tfrac{1}{2}) \right\} = -2.531 \end{aligned}$$

since $\Gamma'(\frac{1}{2}) = \sqrt{\pi}(-\gamma - 2\log 2)$, $\Gamma'(1) = -\gamma$, where γ is Euler's constant and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Thus the entropy of X is $H_X(\theta) = \log(\theta/4\pi)$, obtained without using direct integration. The identity $\int f_Z(z) \log f_Z(z) dz = -\log 4\pi$ follows.

The estimator (4) becomes

$$\Pi_{i=1}^n \left\{ 1 + (x_i/\hat{\theta})^2 \right\} = 4,$$

in a case where the method of moments fails since X has no finite moments. The Cramér-Rao lower bound for unbiased estimates of θ is $2\theta^2/n$; it would be of interest to compare the asymptotic variance of the proposed estimator with this bound.

5.5 Pareto distribution

The density function of this distribution is

$$f(x, k, a) = \frac{ak^a}{x^{a+1}} \quad (x > k; a > 0, k > 0),$$

where the parameters a and k are shape and threshold parameters respectively. If k is known, this is a member of the one-parameter exponential family, but it is not a member of the two-parameter exponential family.

The estimator (3) becomes

$$\bar{g} = k \left\{ 1 + \left(1 + \frac{1}{a} \right) s \right\}^{1/(s(a+1))},$$

As $s \rightarrow 0$, this becomes the maximum likelihood estimator, $\bar{g} = ke^{1/a}$.

Obvious choices of s are $s = \pm 1/(a+1)$ for which the estimator becomes

$$\bar{g} = k \left(1 + \frac{1}{an} \right)^n \quad \text{and} \quad \bar{g} = k \left(1 - \frac{1}{an} \right)^{-n},$$

respectively. Since $E(\bar{g}) = k \left(1 - \frac{1}{an} \right)^{-n}$, the latter is the estimator based on the sample geometric mean. For k known, these yield the estimators

$$\hat{a} = \left[n \{ (\bar{g}/k)^{1/n} - 1 \} \right]^{-1} \quad \text{and} \quad \hat{a}' = \left[n \{ 1 - (\bar{g}/k)^{1/n} \} \right]^{-1},$$

bounding the likelihood solution $\hat{a}_{ml} = \{\log(\bar{g}/k)\}^{-1}$.

6. Concluding remarks

The applicability of this estimation procedure to scale-related parameters in continuous distributions has been demonstrated. Its performance in estimating scale in Cauchy and Pareto distributions hints at the existence of new classes of useful estimators in other long-tailed distributions, not belonging to the

exponential family. The lognormal distribution is an obvious candidate, particularly in view of the recent interest in the options-pricing model of Black-Scholes. It is well known that some of Sichel's models, in particular his model for evaluating diamond deposits, made extensive use of the lognormal law. Sichel (1967) applied his (discrete) 'frequency moment' estimation principle to lognormal samples and demonstrated large sample optimality properties of the estimators thus derived. In view of this, it would be of interest to examine continuous frequency moment estimates in lognormal distributions, as well as in other distributions used in modelling, e.g. the generalised gamma, beta, Weibull, and inverse Gaussian.

There are also some interesting computational possibilities in the proposed procedure. In the case of exponential families, it provides a class of estimators, (4), indexed by s which generalises maximum likelihood. The computational implications are twofold. Firstly, it provides the exact ordinate of the likelihood function at the maximum likelihood estimator, so that any parametric estimator in this family can be checked against this condition for maximising likelihood. The result can also be used to compute the maximum likelihood estimate in any member of a one-parameter exponential family without calculus. Secondly, since the function $\pi_X(s, \theta)$, which defines the class is increasing in s and gives the (negative) entropy as $s \rightarrow 0$, two estimates obtained from (3) with positive and negative s will bound the maximum likelihood estimator with arbitrary accuracy for sufficiently small s -values. Letting $s \propto 1/n^\alpha$ for some $\alpha \geq 1$ generates classes of estimates which approach maximum likelihood (almost surely and in distribution) as $n \rightarrow \infty$.

In distributions not belonging to the exponential family, (3) yields estimators which do not converge to maximum likelihood, providing an alternative approach to the traditional methods of estimation.

7. References

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